ARC Algorithm Metrics & Configurations

This document outlines the specific metrics, hyperparameters, and environment configurations used for both the Baseline 2D ARC system and the Advanced 3D Multi-Agent TD3-ARC system.

1. Baseline 2D ARC (DDPG) Metrics

Environment & State Space

- **Algorithm:** Deep Deterministic Policy Gradient (DDPG)
- **Dimensions:** 500x500 meters (Planar)
- **Action Space:** 2 (Heading, Speed)
- **Sensors:** 9 (2D ray-cast beams)
- **Reward Structure:** Kinematic Shaped Reward (Goal Progress, Heading Alignment, Obstacle Clearance)

Kinematics & Time Metrics

- **UAV Speed (`uav_speed`):** 3.0 m/s
- **Target Speed (`target_speed`):** 0.8 m/s (0.0 m/s for stationary targets)
- **Sampling Interval (`dt`):** 1.0 seconds
- **Max Steps per Episode:** 500

Network Configuration

- **LSTM Hidden Size:** 128
- **Actor/Critic Hidden Size:** 256
- **History Length:** 5 steps

Training Hyperparameters

- **Batch Size:** 32
- **Memory Capacity:** 4,000 (Standard FIFO Replay Buffer)
- **Learning Rate:** 1e-3
- **Exploration Noise:** 0.2
- **Soft Update Rate (τ):** 0.01

2D Reward Shaping Metrics

- **Step Penalty (R_STEP):** -0.02
- **Goal Progress Reward (K_PROGRESS):** 4.0
- **Heading Alignment Bonus (K_ALIGN):** 0.3
- **Obstacle Clearance Penalty (K_CLEARANCE):** 0.8
- **Safety Distance Threshold (D_SAFE):** 0.80

2. Advanced 3D ARC (Multi-Agent TD3 + HER) Metrics

Environment & State Space

- **Algorithm:** Twin Delayed DDPG (TD3) + Hindsight Experience Replay (HER) + Prioritized Experience Replay (PER) + Curriculum Learning
- **Dimensions:** 20x20x15 meters (Dense 3D volume)
- **Action Space:** 3 (Delta Pitch, Delta Yaw, Dynamic Speed)
- **Sensors:** Continuous 3D Signed Distance Fields (SDF)

Kinematics & Time Metrics

- **UAV Speed:** Dynamically controlled by the agent's continuous action space (varies based on energy/progress goals).
- **Wind Disturbance:** Stochastic wind vectors applied per step.
- **Max Steps per Episode:** Scaled dynamically via Curriculum Learning stages.

Network Configuration

- **LSTM Hidden Size:** 256 (2 layers, includes LayerNorm and 0.05 Dropout)
- **History Length:** 10 steps

Training Hyperparameters

- **Batch Size:** 256
- **Memory Capacity:** 100,000
- **Discount Factor (γ):** 0.98
- **Soft Update Rate (τ):** 0.005

TD3 & HER Specific Metrics

- **Policy Delay:** 2 (Actor updates once per 2 Critic updates)
- **Target Noise:** 0.1
- **Target Noise Clip:** 0.3
- **HER Ratio (her_k):** 6 (6 virtual goals generated per 1 real transition)

3D Reward Shaping Metrics

- **Step Penalty (R_STEP):** -0.05
- **Goal Progress Reward (K_PROGRESS):** 4.0
- **Energy Penalty Rate (K_ENERGY):** 0.05
- **Out of Energy Penalty (R_ENERGY_OUT):** -30.0